cs5800-hw_3_exercises

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1 Exercise 3.13

Undirected vs. directed connectivity.

(a) Prove that in any connected undirected graph $G = (V, E)$ there is a vertex $v \in V$ whose removal leaves G connected. (Hint: Consider the DFS search tree for G.)

Answer:
For each connected undirected graph G, there is a DFS search tree T that connects all vertices in G. Because G is connected, T is also connected.

For each tree T, there is at least 1 leaf vertex. Let this vertex be v, then $v \in V$.

Since a leaf vertex in a tree has no children, by removal of v, the new tree T' is a subgraph of T. Because T is connected, any subgraph of T is also connected. So T' is connected.

There is a subgraph $G'$for G whose DFS search tree is T'. because T' is connected, G' is also connected.

Therefore, there is a vertex $v \in V$ whose removal leaves G connected.

(b) Give an example of a strongly connected directed graph $G = (V, E)$ such that, for every $v \in V$, removing v from G leaves a directed graph that is not strongly connected.

Answer:
For a doubly linked list G that has 3 vertices or more, G is considered a strongly connected directed graph. Removing any vertex between the first and the last one will leave G to be not strongly connected.

(c) In an undirected graph with 2 connected components, it is always possible to make the graph connected by adding only one edge. Give an example of a directed graph with two strongly connected components such that no addition of one edge can make the graph strongly connected.

Answer:
For a directed graph G with 2 vertices $v1$ and $v2$ that are not connected, both vertices are considered strongly connected components. There is no way to add only 1 edge to make $v1$ and $v2$ strongly connected. It can be strongly connected by at least adding 2 edges.

2 Exercise in Another PDF

3 Acknowledgements

3.0.1 Resources

Definition of meta-graph, source, and sink, Page 102-104, Dasgupta et. al. text.

3.0.2 LLM Disclosure

4 Reflection

It is hard to remember the property of a graph. By doing the homework, it is slightly easier to remember.